

v3.0

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Content

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v3.0

v3.0 is now [available on NuGet](#), and brings exciting new improvements - and some necessary changes. For most people 3.0 should be a simple update. A few niche features have changed or been removed - more details below. The library tries to guide you at build time if you are particularly impacted.

New Features

- buffer-focused APIs: like many serializers, protobuf-net has historically been focused on the `Stream` API, but recent advances in .NET IO have brought more focus on things like `ReadOnlyMemory<byte>`, `ReadOnlySequence<byte>`, and `IBufferWriter<byte>`; protobuf-net is now able to directly utilize the APIs, to allow use with modern .NET IO primitives without intermediate layers. Since there are performance advantages to these APIs, protobuf-net will also try to detect scenarios where it can use these features - for example, if you try to deserialize from a `MemoryStream`, protobuf-net will use `TryGetBuffer` to access the underlying memory directly when possible
- new serializer core for AOT: the heart of the serialize and deserialize code has been heavily refactored, as part of in-progress work to enable full AOT usage via the proposed “generators” C# language feature; this work is not yet fully available (as the compiler changes are still in preview), but the building blocks are in place and have been validated; AOT usage will be entirely optional, and targets both cold-start performance for “full” runtimes, and ref-emit avoidance for AOT environments like mobile devices and gaming platforms

- new custom serializer API: there is a new API at the heart of protobuf-net that allows custom serializers to be defined in C# - both for messages and primitive values; this API is not yet intended for “consumer” use - that is a later v3 stage - but: it is already opening up a lot of options for the engine, and offers *future* extensibility for consumers
- performance: the use of modern APIs, plus additional changes (things like `ArrayPool<T>` usage) should see performance improvements for most callers, in both serialization and deserialization; note that for 3.0 the only *aim* was “no performance regression” (with a later 3.something deliverable being an in-depth performance review), so getting a performance increase here was a free win!
- pre-measurement of payloads: in a lot of cases, it is necessary to know the *length* of a payload *before* you actually write it - for example, for an HTTP `Content-Length` header; previously this has required serializing the object fully to a buffer, writing the buffer length, then copying the buffer; a new `Measure<T>` API now allows the caller to perform a pre-computation of the length in advance, in a way that allows the library to *reuse* that knowledge in the anticipated serialize that will surely follow, which in turn makes the final serialize a “forwards only” affair
- compatibility level API: this extends the `DataFormat.WellKnown` concept from v2 allowing granular changes to how common data types are serialized; this is discussed [in more depth here](#) - thanks here go to [ServiceStack](#), who both inspired and sponsored these improvements
- capability API: since there are some changes in v3.0 that might make it not a perfect update for everyone, **libraries** that *consume* protobuf-net might not want to take a hard 3.0 dependency, but might still want to *use* the 3.0 features when they are available; new capability APIs make this possible - for example, `protobuf-net.Grpc` uses this, allowing it to use the “contextual serializer” gRPC API when 3.0 APIs are detected, or the legacy `byte[]` API otherwise); this works via interfaces that are *advertised* in the later 2.4 releases, but not *implemented* until 3.0; in particular, you can now test a `TypeModel` for:
 - input - `IProtoInput<T>` where `T` is any of:
 - 2.4 and 3.0: `Stream`, `ArraySegment<byte>`, `byte[]`
 - 3.0 only: `ReadOnlyMemory<byte>`, `ReadOnlySequence<byte>`
 - output - `IProtoOutput<T>` where `T` is any of:
 - 2.4 and 3.0: `Stream`
 - 3.0 only: `IBufferWriter<byte>`
 - output with measurement - `IMeasuredProtoOutput<T>` where `T` is any of:
 - 3.0 only: `Stream`, `IBufferWriter<byte>`
- split libraries: as part of the AOT drive, the library has now been split into `protobuf-net` and `protobuf-net.Core`, with the “Core” package having the main serializer API but none of the runtime reflection code; this means that future AOT-only scenarios only require the “Core” package, and can omit the overhead of shipping any of the other code *at all*; you can see this in action already at [protogen.marcgravell.com](#), which uses the schema

parting tools *and serializer core* in the client browser (not at the web-server) via “Blazor”/WASM, via a dependency on `protobuf-net.Core` (a huge thanks once again to [R emi BOURGAREL](#) here, who leant their Blazor expertise)

Changes To Existing Features

Listed in highest-to-lowest impact; the first 2 here could well be reasons *not* to choose v3 for now

- reference tracking **is deprecated**; this means any use of `AsReference = true`, to build an object *graph* rather than an object *tree*; the reality is: protobuf itself is a tree serializer, and this feature was painful to use and support; I’m *open* to discussions around reinstating it, but it is a *lot* of work, and is a very niche feature that provides a lot of headaches
- dynamic typing **is deprecated**; this means any usage of `DynamicType = true`, where a property is typed as `object`, and the serializer includes the type metadata in the payload; this feature suffers from a number of the [same problems as BinaryFormatter](#), exacerbated by problems due to types moving around between .NET Framework and .NET Core; this made it both brittle and dangerous, in addition to just being *terrible* for cross-platform work; the plan here is to implement support for the `Any` feature of protobuf, which should provide a supportable API, while also allowing compatibility with other protobuf implementations; this work is planned for a future v3 stage, and will be back-ported to v2 to allow a migration route between v2 and v3
- enum maps **are deprecated**; this is where an `enum` has a `[ProtoEnum(Value=...)]` marker that was used to provide a **different** value for serialization - for example, when the enum value is `42` but you want to serialize it as `3`; this feature just confused a lot of people and made code brittle; instead, it is suggested to use a shadow property **in your code** that provides this map, for example via a `switch` expression, and serialize the shadow property instead
- `ProtoReader / ProtoWriter` must now be instantiated via `ProtoReader.Create`, not `new` (recent v2 releases have been advising this change for some time); most users will not be using these APIs
- minor: `ProtoReader / ProtoWriter` *should* now be used with the `ref state` APIs when used against `Stream`, and *must* now be used with the `ref state` APIs when used against the new buffer APIs; most users will not be using these APIs
- minor: `DataFormat.WellKnown` now advises you to use `CompatibilityLevel` instead; if you’re happy with your code “as is”, you can safely ignore this warning (`#pragma` or similar), and your code will continue to work in `Level240` (as [discussed here](#))
- minor: the API for creating runtime type models has moved from `TypeModel.Create` to `RuntimeTypeModel.Create`; this is largely a side-effect of the library split, and recent v2 releases have been advising this change for some time; if this causes genuine inconvenience to anyone, we can probably “fix” this - just ping me!
- streamlined framework targets; we used to have targets for (takes a deep breath): .NET Framework 2.0/3.5/4.0, .NET Standard 1.0/1.3/2.0, UAP 10.0, and .NET Core 2.1; carrying

around this weight made feature changes hard bordering on impossible, and in reality: I wasn't happy that we were providing suitable testing for platforms that *we haven't used in years*; the TFM list is now .NET Standard 2.0/2.1, and .NET Framework 4.6.1; for more discussion on this theme, [see my writing here](#)

Summary

Emphasis: for most people, this should be a simple upgrade that “just works”. It has been in use at Stack Overflow for a very long time, without any problems - providing new performance and features. If you are using the reference tracking or dynamic type features, or .NET Framework 3.0, then I totally get that you might be frustrated here. The good news is that *nobody is coming to uninstall v2 from your system* - it will continue to work just fine.

As always, this library is provided “as is” etc, but if you have any problems or find any bugs, please [let me know](#) and we'll try to fix it.

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